# MODULE 7 — Human–AI Collaboration in Data Science

### Co-Design, Not Automation

## Why This Module Matters

Up to now, the lesson plan has focused on:

- **what decisions are being made**
- **how pipelines shape uncertainty**
- **how failures emerge end-to-end**

This module answers a different question:

**Where should AI help think — and where must humans stay in control?**

Most real-world failures with AI-assisted data science come from **over-delegation**, not bad intent.

## Core Reframe (Memorize This)

**AI is a collaborator in reasoning — not an authority on decisions.**

Strong systems are **co-designed**:

- AI proposes
- AI critiques
- Humans decide

## Roles AI Can Play (and Why This Matters)

AI is not "on" or "off". It plays **different roles at different pipeline stages**.

### 1️⃣ AI as Analyst

- generates options
- enumerates assumptions
- surfaces tradeoffs
- proposes alternatives

✅ High leverage

⚠️ Must be constrained

### 2️⃣ AI as Reviewer

- critiques proposed pipelines
- identifies failure modes
- stress-tests assumptions
- challenges confidence

✅ Safest role
✅ Underused in practice

### 3️⃣ AI as Implementer

- writes code
- applies transformations
- executes defined plans

⚠️ Most dangerous role
⚠️ Requires tight guardrails

## Safe vs Dangerous Automation

### ✅ Safe to Delegate to AI

- listing assumptions
- proposing multiple strategies
- identifying risks and edge cases
- generating draft artifacts
- critiquing its *own* outputs

### ❌ Dangerous to Delegate to AI

- defining success metrics alone
- choosing thresholds autonomously
- deciding when evidence is “good enough”
- silently patching pipeline failures
- approving deployment readiness

Automation without review is **risk amplification**.

## Pipelines as DAGs (Not Scripts)

AI-native collaboration treats pipelines as **graphs**, not notebooks.
Each node represents:

- an assumption
- a risk boundary
- a decision commitment

Edges represent:

- dependency
- constraint propagation
- failure amplification paths

This is why **review checkpoints** matter.

## Human Review Checkpoints (Critical)

Humans must explicitly review at:

- decision framing
- data trust boundaries
- cleaning strategy choice
- feature inclusion
- model class selection
- metric & threshold definition

If humans only review *outputs*, they are too late.

## Intermediate Artifacts (Why They Exist)

Artifacts are not bureaucracy — they are **thinking checkpoints**.
Examples:

- decision statements
- assumption lists
- rejected alternatives
- pipeline DAGs
- failure narratives

AI helps produce them. Humans validate and own them.

## Prompting Pattern for Co-Design

Instead of:
"Build me a churn model."
Use:
**"Act as a senior data scientist reviewing this pipeline. List assumptions, risks, and failure modes before suggesting changes."**
This keeps AI in a **reviewer role**, not a decision-maker role.

Appendix

- Under Module 7, see *Examples of How To Develop Prompts for each Data Science Pipeline Stage*
- Under Module 7, see *Examples of How to Develop Sprint-Friendly Prompts for each Data Science Pipeline Stage*

## 🧪 Module 7 Mini-Exercise — Mock AI Design Review

This exercise practices **safe human–AI collaboration**.

### Step 1 — Provide the Pipeline Context

Briefly restate:

- the decision being supported
- the high-level pipeline structure
- the strategy being used (A or B)

### Step 2 — Ask AI to Critique the Pipeline

Write the prompt you would give AI to:

- review the pipeline
- identify assumptions
- list risks
- flag dangerous automation points

*(You are not running the prompt — just writing it.)*

### Step 3 — Identify Mandatory Human Review Points

List **3 pipeline stages** where:

- AI suggestions must be reviewed
- humans retain final authority

Explain *why* each checkpoint is human-owned.

### Step 4 — Explicit Delegation Boundary

Complete this sentence:

**"AI is allowed to ___, but must never ___ in this system."**

Be concrete.

## Rules (Important)

❌ No production tooling
❌ No governance frameworks
❌ No deployment talk
✅ Focus on collaboration roles
✅ Explicit human responsibility
✅ Clear delegation boundaries

## Why This Module Belongs *Here*

Module 7 ensures that:

- AI accelerates thinking without replacing judgment
- humans stay accountable
- pipelines remain interpretable and defensible

This is the **last step before readiness discussions**.

## Relevant Articles

*Human-AI Collaboration*

- https://thedecisionlab.com/reference-guide/computer-science/human-ai-collaboration?utm_source=chatgpt.com

*How to Build Your Human-In-the-Loop Pipeline: A Step-by-Step Guide*

- https://humansintheloop.org/how-to-build-your-human-in-the-loop-pipeline-a-step-by-step-guide/?utm_source=chatgpt.com

*How to Supercharge Your Design Workflow with AI*

- https://www.interaction-design.org/literature/article/how-to-supercharge-your-design-workflow-with-ai?srsltid=AfmBOor8KqU5qAdRoOa30PAt4QdEN_fxH-Nu4xcRldSSHQd0bWatuyyd&utm_source=chatgpt.com

*Automate Amazon SageMaker Pipelines DAG Creation*

- https://aws.amazon.com/blogs/machine-learning/automate-amazon-sagemaker-pipelines-dag-creation/?utm_source=chatgpt.com